

Education

- 2018–2023 **PhD in Computer Science** Oregon State University, Corvallis, USA
- Research: Explainability of Computation, Designing Domain-Specific Languages (DSLs)
 - Thesis: *Automatic Explanations of Computation Results with Value Decompositions and Dominating Sets*
- 2014–2017 **MSc in Computer Science** University of Calgary, Calgary, Canada
- Research: Designed and implemented Message Passing Language (MPL), a type-safe concurrent programming language
 - Thesis: *Implementation of Message Passing Language*
- 2007–2011 **B.Tech in Computer Science** Cochin University of Science & Technology, Cochin, India

Academic Experience

- Aug 2023–Present **Postdoctoral Researcher** Johannes Gutenberg University Mainz, Germany
- Designed and taught *Advanced Topics in Functional Programming* (Fall 2023, 2024)
 - Teaching Assistant for the Program Analysis course (Winter 2024)
 - Supervised two undergraduate Bachelor's theses
 - Research in programming languages & formal verification methods
 - Assisted in reviewing papers for GPCE (2023), ECOOP (2024, 2025)
- 2018–2023 **Research & Teaching Assistant** Oregon State University, USA
- Conducted research on computational explanations under DARPA's eXplainable AI (XAI) program
 - Assisted in reviewing papers for POPL (2018), GPCE (2021), and VLHCC (2020, 2021)
 - Secured NSF Research Experiences for Undergraduates (REU) grant (\$8,000) for undergraduate research on explanation visualization.
 - Teaching Assistant for Programming Paradigms (CS 381, 1 Semester)
 - Mentored 3 high school students in functional programming (Haskell, Elm) under Apprenticeships in Science and Engineering (ASE) program
- 2014–2018 **Research & Teaching Assistant** University of Calgary, Canada
- Teaching Assistant for the following courses:
 - Compiler Construction (CPSC 411, 4 semesters)
 - Foundations of Functional Programming (CPSC 521, 2 semesters)
 - Programming Paradigms (CPSC 449, 3 semesters)
 - Developed MPL compiler with novel aspects being its type system implementation as well as an abstract machine for executing MPL programs

Professional Experience

- 2011–2013 **Software Engineer** Accenture, Mumbai, India
- Implemented SAP HCM Payroll systems and developed ITIL/SOX-compliant solutions
 - Led system configuration, testing, and root cause analysis for enterprise projects

Technical Skills

Programming Haskell (Expert), Scala, Prolog, Coq, Python, LaTeX
Skills Compiler Construction, Abstract Machine Design, Domain-Specific Language (DSL) Design

Languages

Proficiency English (Fluent), Hindi (Native)

Teaching Expertise

Core Courses Programming Paradigms, Compiler Construction
Advanced Courses Advanced Functional Programming, Program Analysis, DSL Design

Publications

Journal Articles

- 2024 Erwig, M., Kumar, P. *Explanations for Combinatorial Optimization Problems* – Journal of Computer Languages, Vol. 79
- 2021 Erwig, M., Kumar, P. *Explainable Dynamic Programming* – Journal of Functional Programming, Vol. 31, No. e10

Conference Publications

- 2024 Pacak, A., Kumar, P., Xu, R., Erdweg, S. *AUTOINC: Incrementality for Free* – SPLASH Companion '24 (**Short Paper**)
- 2023 Kumar, P., Erwig, M. *MatchMaker: A DSL for Game-Theoretic Matching* – Symposium on Trends in Functional Programming, LNCS 13868 (**John McCarthy Best Paper Award**)
- 2021 Erwig, M., Kumar, P. *MADMAX: A DSL for Explanatory Decision Making* – ACM SIGPLAN Conference on Generative Programming: Concepts & Experiences
- 2020 Erwig, M., Kumar, P., Fern, A. *Explanations for Dynamic Programming* – International Symposium on Practical Aspects of Declarative Languages, LNCS 12007

Under Progress

- 2025 Kumar, P., Pacak, A., Erdweg, S. *Incremental Computing by Differential Execution* – **Submitted to ECOOP, 2025**
- 2024 Erwig, M., Kumar, P. *Explaining Results of Multi-Criteria Decision Making* – Journal of Multi-Criteria Decision Analysis **In Revision (Minor Revision)**

Thesis

- 2023 Kumar, P. *Automatic Explanations of Computation Results with Value Decompositions and Dominating Sets* – PhD Thesis
- 2018 Kumar, P. *Implementation of Message Passing Language* – MSc Thesis

Selected Talks

Conference Presentations

- 2023 *Design of a DSL for Game-Theoretic Matching* – Trends in Functional Programming, University of Massachusetts, Boston
- 2021 *A DSL for Explanatory Decision Making* – Generative Programming: Concepts & Experiences (GPCE), Chicago
- 2020 *Explanations for Dynamic Programming* – Practical Aspects of Declarative Languages (PADL), New Orleans

Invited Talks

- 2022 *MPL and Explainability of Computation* – Tech Talk at Galois, Portland
- 2018 *Message Passing Language: Design and Implementation* – Departmental Seminar, University of Calgary

Workshop Presentations

- 2017 *Message Passing Language: Design and Implementation* – FMCS, University of Ottawa
- 2016 *Abstract Machine for Message Passing Language* – FMCS, University of British Columbia
- 2014 *Survey of Abstract Machines for Programming Languages* – FMCS, University of Calgary